

LEDGER · 2026-08-30 · MATERIALS SCIENCE / THERMAL ENERGY STORAGE

Form-Stable Capric-Palmitic/Expanded Graphite (CA-PA/EG) Composites via Vacuum Impregnation

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	PureTemp 20
Entry application	High-performance, multi-day pharmaceutical cold-chain packaging compliant with ISTA 7D thermal profiles for Controlled Room Temperature (CRT, 15°C–25°C) biologics transit
Measured advantage	>10.0x faster thermal response (>2.0 W/mK vs 0.14-0.23 W/mK)
Time to first revenue	6 months
Gross margin at parity	41.6%
Startup CapEx	\$44,000
Payback from first sale	4.6 months
Capital productivity	10.64x

Why it failed

- Voided by: Economics undisclosed, Form-factor / handling regression
- Margin fails the operating-cost check

Assessment

The concept fails the 70% gross margin threshold, delivering only 42.6% gross margin at parity pricing (\$6.50/kg) against the incumbent PureTemp 20. It also introduces severe form-factor regressions with black graphite powder handling hazards, compression weeping, and latent heat loss.

Also flagged

- Evidence rests on non-peer-reviewed sources
- Duplicate references inflating the evidence base
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-30 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

